

ECON 3406: Statistics for Business II (Summer 2026)

General Course Information

Instructor:	Dr. Joshua Martin	Email:	joshua@westga.edu
Office:	338 Roy Richards Sr. Hall	Office Hours:	MW 3:00-5:00pm, or by apt.
Course Delivery:	Online	Section:	E01

Prerequisite

(ECON 3402 or MATH 1401) and MATH 1413

Course Description

This course introduces intermediate quantitative tools for strategic and data-driven decision-making in business. Topics include probability, hypothesis testing, statistical inference, linear regression, multiple regression, logistic regression, and simulation analysis.

In addition to developing a solid foundation in statistical theory, students will gain hands-on experience using *R*, an open-source statistical programming language widely adopted in both industry and research. Emphasis is placed on interpreting statistical results, visualizing and analyzing real-world data, and communicating findings effectively.

Course Textbook

David M. Diez, Mine Çetinkaya-Rundel, and Christopher D. Barr. [OpenIntro Statistics](#) (Fourth Edition)

Course Goals

The learning goals for this course are as follows:

1. Recognize the importance of data collection, identify limitations in data collection methods and other sources of statistical bias, and determine their implications and how they affect the scope of inference.
2. Use statistical software to summarize data numerically and visually, and to perform data analysis.
3. Have a conceptual understanding of the unified nature of statistical inference.
4. Apply estimation and testing methods to analyze single variables or the relationship between two variables in order to understand natural phenomena and make data-based decisions.
5. Model numerical response variables using a single explanatory variable or multiple explanatory variables in order to investigate relationships between variables.
6. Interpret results correctly, effectively, and in context without relying on statistical jargon.
7. Critique data-based claims and evaluate data-based decisions.

Technology Requirements

As part of this course, we will need reliable access to the internet and technology to be successful. The Ingram Library on the Carrollton campus has public computers with internet access available. The UWG Newnan campus has public computers with internet access available for students. Minimum Technology Requirements include: Reliable access to a PC, Mac, Chromebook, or tablet; A webcam with speakers or headphones; High speed internet connection; Up-to-date Chrome, Firefox, or Safari browsers; CourseDen access; Microsoft Word or Google Docs access.

Technology Support and Guidance Available for Students: [TechWest Student ITS Support](#); [UWG Online](#); [24/7 CourseDen \(Brightspace D2L\) support](#)

Evaluation Policies

- Exams are closed notes and must be completed using the Respondus LockDown Browser within CourseDen.
 - Assignments are time-limited to 3 minutes per question.
- R Labs are open-note and may be completed collaboratively with your classmates and/or with the help of LLM (“artificial intelligence”) models.
 - The purpose of these assignments is to gain deeper insights into data analysis and statistics through learning a powerful, flexible statistical programming language rather than seeking “perfect” answers.
 - Thus, clearly copied code from another student or LLM model, will be heavily penalized.
 - Ideally, for the sake of simplicity, coding styles should entirely be in base R and should mirror those seen in the notes.
- Make-up Policy:
 - There are no make-ups for missed assignments.
 - Assignments missed due to an excused absence will be replaced with the average of the remaining assignment scores.
 - Excused absences require documentation from a medical professional or relevant university official.
- Academic Integrity:
 - Any *suspected* academic dishonesty will result in an immediate zero on the assignment.
 - A second instance will result in immediate course failure and referral for further disciplinary action.

Grade Scale

Grade	Range
A	[90,100+)
B	[80, 90)
C	[70, 80)
D	[60, 70)
F	[0, 60)

Grading

- **R Labs**
 - There will be 9 R Lab, each worth 3 points.
 - The rubric for these assignments can be seen below.

Grade	Description
3	Outstanding
2.5	Great
2	Improve
1.5	Incorrect
1	Effort Shown
0.5	Submitted
0	Missing

- **Exams**
 - There will be 4 exams and a comprehensive final, each worth 15 points.
 - If $Exam_5 > 0.6$, it may replace your lowest exam score; otherwise, it counts as any other exam.

$$Exams = \begin{cases} 15 \times \sum_{e=1}^5 Exam_e & \text{if } Exam_5 \leq 0.6, \\ 15 \times \left(\sum_{e=1}^4 Exam_e - \min(Exam_e) \right) + 30 \times Exam_5 & \text{if } Exam_5 > 0.6. \end{cases}$$

- **Exam Curves**

- I write exams to be challenging, so I apply a consistent curve to keep the grading scale fair and to avoid “extra credit-style” outcomes where top scores vastly exceed 100.
- Let g denote a student’s raw exam score out of 100 points.
- A constant bonus c (in points) is added to every raw score where $c \geq 0$ is the smallest integer such that *at least one* of the following stopping conditions is met across the class distribution:
 - * The mean boost satisfies $f - g \geq 25$,
 - * The class mean final grade satisfies $\bar{f} \geq 80$, or
 - * At least one student’s final grade reaches 100.

$$f(g) = \frac{c}{2} + \frac{g+c}{100} \times \left(100 - \frac{c}{2}\right)$$

Course Schedule

Date	Day	Readings	(Target) Due Date
06/29/26	Mon	Ch. 1.1-1.3	
06/30/26	Tue	Ch. 1.3-1.4	RLab 1
07/01/26	Wed	Ch. 2.1	RLab 2
07/02/26	Thu	Ch. 2.2-2.3	Exam 1
07/03/26	Fri	Ch. 3.1-3.2	
07/05/26	Sun		Unit 1 Due
07/06/26	Mon	Ch. 4.1-4.3	RLab 3
07/07/26	Tue	Ch. 5.1-5.2	RLab 4
07/08/26	Wed	Ch. 5.3	RLab 5
07/09/26	Thu		Exam 2
07/10/26	Fri	Ch. 6.1-6.2	
07/12/26	Sun		Unit 2 Due
07/13/26	Mon	Ch. 6.3-6.4	RLab 6
07/14/26	Tue	Ch. 7.1-7.3	
07/15/26	Wed	Ch. 7.4-7.5	RLab 7
07/16/26	Thu		Exam 3
07/17/26	Fri	Ch. 8.1-8.2	
07/19/26	Sun		Unit 3 Due
07/20/26	Mon	Ch. 8.3-8.4	RLab 8
07/21/26	Tue	Ch. 9.1-9.2	RLab 9
07/22/26	Wed	Ch. 9.3-9.5	Exam 4 / Unit 4 Due
07/23/26	Thu		Exam 5

Artificial Intelligence Use Policy

Use of Gen AI tools is not permitted for any graded assessments in this course. Submitting work that is entirely or largely generated by AI will be treated as plagiarism and handled in accordance with the academic and disciplinary policies outlined in the UWG Honor Code (see: [Student Handbook](#)).

That said, Gen AI tools – such as OpenAI’s ChatGPT – will be incorporated in class activities to support learning. For example, I may use AI to generate commented versions of code demonstrated in class, explaining each function and its purpose. This use is intended to help you focus on understanding statistical theory while still gaining confidence with coding concepts and syntax.

Center for Academic Success

The [Center for Academic Success](#) provides services, programs, and opportunities including academic coaching, peer tutoring, drop-in tutoring and supplemental instruction to help all undergraduate students succeed academically.

For more information, contact them: (678) 839-6280 or cas@westga.edu.

Accessibility Services

Students with a documented disability may work with UWG Accessibility Services to receive essential services specific to their disability. All entitlements to accommodations are based on documentation and USG Board of Regents standards.

If a student needs course adaptations or accommodations because of a disability or chronic illness, or if they need to make special arrangements in case the building must be evacuated, the student should notify their instructor in writing and provide a copy of their Student Accommodations Report (SAR), which is available only from Accessibility Services. Faculty cannot offer accommodations without timely receipt of the SAR; further, no retroactive accommodations will be given.

For more information, please contact [Accessibility Services](#).

Mental Health Support

If you or another student are experiencing a mental health issue, free and confidential services are available through the [Counseling Center](#). Students who have experienced sexual or domestic violence can access confidential support through Patient Advocates in [Health Services](#). To report a concern anonymously, please visit [UWG Cares](#).

English Language Learning (ELL) Resources

If you are having difficulty with English or U.S. academic culture, specialized resources are available to help you succeed. Visit the [ELL Resource Page](#) for more information.

Honor Code

At the University of West Georgia, we believe that academic and personal integrity are based upon honesty, trust, fairness, respect, and responsibility.

Students at West Georgia assume responsibility for upholding the honor code. Students pledge to refrain from engaging in acts that violate academic and personal integrity. These include, but are not limited to, plagiarism, cheating, fabrication, aiding academic dishonesty, lying, bribery or threats, and stealing.

Academic Dishonesty Tracking System

UWG maintains and monitors a confidential Academic Dishonesty Tracking System that collects and reports patterns of repeated violations across all colleges, the Ingram Library, and the School of Nursing. Each incident is reviewed by the instructor and may result in academic penalties including, but not limited to, failing the assignment or the course. Conduct sanctions may range from a warning to suspension or expulsion.

Students are responsible for safeguarding their computer accounts. All actions originating from their account or network connection are their responsibility. Impersonation or misrepresentation of identity is prohibited.

For more details, see the [Student Handbook](#).

Credit Hour Policy

UWG grants one semester hour of credit for work equivalent to a minimum of one hour (50 minutes) of in-class instruction and two hours of out-of-class student work per week for approximately fifteen weeks.

Each syllabus must document the in-class and out-of-class requirements to earn the assigned credit hours. Out-of-class work may include assignments, readings, observations, or musical practice.

Credit may also be granted via competency-based testing (e.g., AP, CLEP, or departmental exams) that demonstrates mastery of course outcomes.

Changes to the Syllabus

I reserve the right to modify the course schedule or content as necessary to support the learning objectives of the class. Any such changes will be communicated in advance and will only be made in ways that are beneficial to students, such as reducing the scope of material covered or delaying assignment due dates.

Title IX & Equal Opportunity

University of West Georgia faculty are committed to helping create a safe and open learning environment for all students. If you (or someone you know) have experienced any form of sexual misconduct, including sexual assault, dating or domestic violence, or stalking, know that help and support are available. The University strongly encourages all members of the community to take action, seek support, and report incidents of sexual misconduct to the [Office of Title IX & Equal Opportunity](#). Please be aware that under Title IX of the Education Amendments of 1972, I am required to disclose information about such misconduct to the Office of Title IX & Equal Opportunity. If you wish to speak to a confidential employee who does not have this reporting responsibility, you can contact the University of West Georgia Counseling Center (678-839-6428) or Advocate Services (470-215-9946).

Preferred Pronouns Statement

At UWG everyone has the right to be addressed by the name and personal pronouns that correspond to their gender identity, including non-binary pronouns, for example: they/them/theirs, ze/zir/zirs, etc. As of 2023, students can update their pronouns in their Self-Service Banner user settings. If you have not yet updated your pronouns in Self-Service Banner, you can do so at the beginning of the term so that I can make sure to refer to you using the correct pronouns. If your pronoun set is not available in Self-Service Banner/CourseDen, please let me know. I recognize that preferred names and pronouns may change during the semester, if at any point during the semester you would like to be addressed differently, please let me know.